		equilibrium, both objects A and B are at the same temperature.
13	**C**	$Pt = ml_f$ $(50)(8.0)(60) = (2.0)\ l_f$ $l_f = 12000\ \text{J kg}^{-1}$ At 5 min, substance is still solid as it is the start of the melting process. At 15 min, substance has completely melted so it is all liquid as it has not reached boiling point yet. $Pt = mc\Delta\theta$ $\Delta\theta = \frac{P}{mc}t$ Greater value of *c*, smaller gradient which is solid hence specific heat capacity of solid greater than that of liquid.
14	**D**	For large amplitude vibration (resonance), natural frequency must be equal to